

FretGarden Pedagogical Constitution

PEDAGOGICAL STANDARD

Chapter 8 - Feedback and Error Classification	Version 1.0
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This document forms part of the FretGarden Pedagogical Constitution. Normative terms are used according to the Foundation.

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Chapter 8 - Feedback and Error Classification

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1. Purpose

This chapter defines how FretGarden interprets learner errors and how it communicates information intended to improve the next attempt, guide review, support remediation, and preserve honest understanding of progress.

Feedback is not decoration, encouragement, or a statement of correctness alone. It is an instructional intervention. Its quality must be judged by whether it helps the learner understand the relationship between an attempt and the intended objective, select an appropriate correction, and produce stronger evidence in a subsequent attempt.

Error classification is the disciplined interpretation of what an unsuccessful, incomplete, unstable, or inefficient attempt may reveal. The purpose of classification is not to assign labels to learners. It is to distinguish among different causes that require different educational responses.

A wrong note caused by inaccurate note recall is not pedagogically equivalent to a wrong note caused by a rhythmic displacement, an unclear diagram, a physical transition failure, a misunderstanding of the root, or careless continuation at an inappropriate tempo. Treating all errors as equivalent produces generic feedback and ineffective remediation.

This chapter therefore establishes:

- 1 the constitutional purposes of feedback;
- 2 the boundaries of defensible error inference;
- 3 a common error taxonomy;
- 4 rules for feedback timing, specificity, tone, and scope;
- 5 correction and reattempt requirements;
- 6 escalation from local feedback to remediation;
- 7 treatment of uncertainty, partial success, and self-correction;
- 8 standards for learner-facing progress language;
- 9 and conformance requirements for future lessons, exercises, assessments, and adaptive systems.

2. Constitutional Authority

This chapter operationalizes the following commitments:

- 1 feedback **MUST** change or inform the next attempt;
- 2 errors **MUST** be treated as evidence rather than moral failure;

- 3 productive error MUST be distinguished from careless repetition;
- 4 assessment claims MUST not exceed available evidence;
- 5 feedback MUST remain specific, accurate, respectful, and actionable;
- 6 repeated error patterns MUST trigger a change in instructional response;
- 7 FretGarden MUST distinguish missing evidence from negative evidence;
- 8 correction MUST preserve the target objective rather than merely reveal the answer;
- 9 learner-facing language MUST remain honest about current capability;
- 10 and feedback MUST support increasing independence rather than permanent dependence on external correction.

Where another specification treats feedback only as praise, punishment, score display, answer revelation, or interface reaction, this chapter governs.

3. Governing Principle

Feedback exists to reduce the distance between the learner's current attempt and the intended capability while preserving enough learner responsibility for learning to occur.

The governing sequence is:

- 1 identify the objective and relevant quality dimensions;
- 2 observe the attempt under known conditions;
- 3 distinguish success, partial success, error, and missing evidence;
- 4 estimate the most defensible error class without overstating certainty;
- 5 identify the smallest useful correction;
- 6 communicate that correction in precise and respectful language;
- 7 require a new attempt, reconstruction, or explanation;
- 8 observe whether the correction resolved the problem;
- 9 escalate support only when necessary;
- 10 and update the evidence record according to what the corrected performance actually demonstrates.

Feedback that does not influence a subsequent action may still inform the learner, but it does not satisfy the full instructional function required for core practice.

4. Core Distinctions

4.1 Error Is Not Failure of Character

An error is a mismatch between an attempt and the target under specified conditions. It provides information about capability, task design, support, attention, or context. It does not establish laziness, lack of talent, or lack of potential.

4.2 Incorrect Is Not Diagnosed

Knowing that a response is wrong does not establish why it is wrong. Diagnostic language MUST be proportionate to the available evidence.

4.3 Feedback Is Not Answer Revelation

Displaying the correct answer may end uncertainty, but it may also eliminate retrieval and conceal the cause of error. Correction should reveal only the information necessary to restore productive work.

4.4 Praise Is Not Feedback

Praise may support motivation when accurate and specific. It does not independently explain what succeeded or what to repeat.

4.5 Score Is Not Feedback

A score summarizes performance. It is not actionable unless connected to objective dimensions, error patterns, and a next step.

4.6 Correction Is Not Mastery

A correct response immediately after the answer has been shown demonstrates corrected performance under support. It does not establish independent recall, retention, or transfer.

4.7 Self-Correction Is Distinct Evidence

A learner who detects and repairs an error demonstrates more capability than a learner who remains unaware, but less than a learner who performs correctly and stably without correction. Evidence records **SHOULD** preserve this distinction.

4.8 Silence Is Not Neutral

When a learner repeatedly practices an error without feedback or self-detection, the system may strengthen the wrong behavior. Delayed feedback is acceptable only when delay serves a defined educational purpose.

5. Definitions

5.1 Feedback

Feedback is information about the relationship between an attempt and an intended objective that can guide reinforcement, correction, strategy selection, or future practice.

5.2 Error

An error is an observable mismatch between an attempted response and the defined target under specified conditions.

5.3 Error Class

An error class is a pedagogically meaningful category describing the most likely capability, process, or condition associated with an error.

5.4 Error Cause

An error cause is the underlying reason an error occurred. Causes may be inferred but are rarely known with certainty from one attempt.

5.5 Correction

Correction is the process by which the learner identifies, reconstructs, or performs a more accurate response.

5.6 Reattempt

A reattempt is a new opportunity to perform the target after feedback or adjustment.

5.7 Hint

A hint is limited support that narrows the problem without supplying the complete answer.

5.8 Prompt

A prompt is information that initiates or directs a response. Prompts vary in how much solution-relevant information they provide.

5.9 Explanation

An explanation identifies a relationship, rule, or reason intended to improve understanding rather than only produce one correct answer.

5.10 Remediation Trigger

A remediation trigger is evidence indicating that local feedback is insufficient and that targeted re-instruction or prerequisite repair is required.

5.11 Careless Error

A careless error is an error arising primarily from insufficient attention, inappropriate continuation, rushed response, or failure to apply a capability that current evidence suggests is otherwise available.

5.12 Productive Error

A productive error is an incorrect attempt that is relevant to the target, yields interpretable evidence, and is followed by meaningful correction.

6. Purposes of Feedback

Every feedback response **MUST** serve one or more declared purposes.

6.1 Confirmation

Identify what was correct so the learner knows what should be preserved.

6.2 Discrepancy Identification

Identify the dimension in which the attempt differed from the target.

6.3 Error Localization

Direct attention to the note, beat, movement, relationship, step, or decision where the problem occurred.

6.4 Strategy Guidance

Recommend an action such as slowing down, counting, isolating, reconstructing, comparing, listening, or reducing scope.

6.5 Conceptual Repair

Correct an inaccurate model or definition.

6.6 Retrieval Support

Provide a cue that permits the learner to reconstruct an answer without complete revelation.

6.7 Calibration

Help the learner compare self-evaluation with observed performance.

6.8 Evidence Interpretation

Explain what the attempt does and does not demonstrate.

6.9 Progress Guidance

Identify whether the learner should repeat, vary, review, apply, consolidate, or remediate.

6.10 Independence Development

Teach the learner to detect and correct similar errors without external support.

7. Feedback Quality Standard

A feedback response SHOULD be:

- 1 accurate;
- 2 relevant to the objective;
- 3 specific to the observed evidence;
- 4 proportionate to the error;
- 5 concise enough to support action;
- 6 actionable;
- 7 respectful;
- 8 timed appropriately;
- 9 limited in certainty when the cause is uncertain;
- 10 and followed by an opportunity to respond.

Feedback MUST NOT introduce unrelated corrections merely because they are visible. When several dimensions are weak, feedback should prioritize the issue that most limits the target objective or prevents productive continuation.

8. The Feedback Contract

Every core exercise, assessment, or guided practice activity MUST define a feedback contract containing:

- 1 the objective and assessed dimensions;
- 2 the evidence available to the feedback process;
- 3 the error classes that can be distinguished validly;
- 4 the classes that cannot be inferred;
- 5 the timing of feedback;

- 6 the amount of answer information permitted;
- 7 the correction action expected from the learner;
- 8 the number or conditions of reattempts;
- 9 escalation rules;
- 10 and the effect of corrected performance on evidence status.

A feedback system **MUST NOT** claim to diagnose causes it cannot observe.

9. Evidence Before Interpretation

Error interpretation begins with observation.

Relevant evidence may include:

- the learner's response;
- response latency;
- prior attempts;
- prompting level;
- selected answer path;
- timing relationship;
- physical performance where observable;
- explanation or reasoning;
- self-evaluation;
- recent review history;
- and performance under reduced difficulty.

The interpretation **MUST** account for task validity. An unclear prompt, inaccessible response mode, technical malfunction, ambiguous example, or misleading representation may invalidate the error as evidence of learner capability.

10. Error Classification Principles

10.1 Classification Is Objective-Relative

The same response may represent different errors under different objectives.

10.2 Classification Is Provisional

One attempt often supports only a tentative classification. Repeated or contrastive evidence may strengthen confidence.

10.3 Classification Must Be Actionable

A category is useful only when it changes feedback, practice, assessment, or remediation.

10.4 Classification Must Avoid Excessive Granularity

Categories should be detailed enough to guide instruction but not so narrow that evidence cannot distinguish them reliably.

10.5 Multiple Errors May Coexist

A learner may produce correct pitches with incorrect rhythm, or use correct reasoning with unstable execution. Evidence SHOULD remain dimensional.

10.6 Primary and Secondary Error Classes

When several errors occur, the system SHOULD identify the primary error that most directly caused failure and MAY record secondary errors when educationally useful.

11. Error Taxonomy Overview

FretGarden recognizes the following major error families:

- 1 orientation and representation errors;
- 2 perceptual errors;
- 3 retrieval errors;
- 4 conceptual errors;
- 5 fretboard-navigation errors;
- 6 physical-execution errors;
- 7 temporal-control errors;
- 8 musical-application errors;
- 9 metacognitive-practice errors;
- 10 attention and regulation errors;
- 11 transfer errors;
- 12 and task or system errors.

These families correspond to the objective domains defined in Chapter 2 while preserving additional categories required for valid interpretation.

12. Orientation and Representation Errors

An orientation or representation error occurs when the learner misunderstands the task, symbols, diagram, terminology, response method, or region of the instrument rather than the target musical relationship itself.

Examples include:

- confusing string number with finger number;
- reading a fretboard diagram in the wrong direction;
- misunderstanding which note value receives the metronome click;
- interpreting a root marker as a required finger;

- or answering in note names when scale degrees were requested.

The first response should clarify the representation and then retest the musical capability. The original attempt SHOULD NOT be treated as strong negative evidence about the underlying musical objective.

13. Perceptual Errors

A perceptual error occurs when the learner does not detect, discriminate, identify, or anticipate the relevant audible feature.

Subclasses include:

- 1 non-detection;
- 2 confusion between close alternatives;
- 3 context dependence;
- 4 register or timbre dependence;
- 5 tonal-center loss;
- 6 rhythmic masking;
- 7 and mismatch between heard and named response.

Feedback should direct attention to an audible contrast, reduce irrelevant complexity, and require renewed listening. Visual answer revelation alone does not repair perception.

14. Retrieval Errors

A retrieval error occurs when knowledge is not available at the required support level or response condition.

Subclasses include:

- 1 complete recall failure;
- 2 slow but accurate reconstruction;
- 3 cue dependence;
- 4 interference between related items;
- 5 order dependence;
- 6 recognition without recall;
- 7 and temporary lapse after prior mastery.

Feedback should first provide the smallest useful cue. If the learner reconstructs successfully, the task should be attempted again with the cue reduced or removed.

15. Conceptual Errors

A conceptual error reflects an inaccurate or incomplete model of a musical relationship.

Subclasses include:

- 1 definition error;

- 2 category confusion;
- 3 root-tonic confusion;
- 4 interval-counting error;
- 5 formula error;
- 6 causal misunderstanding;
- 7 false generalization;
- 8 exception treated as rule;
- 9 and memorized answer without relational reasoning.

Conceptual errors require explanation, contrast, derivation, or reconstruction. Repeating the same item without repairing the model is insufficient.

16. Fretboard-Navigation Errors

A fretboard-navigation error occurs when the learner cannot locate or connect musical relationships spatially on the instrument.

Subclasses include:

- 1 lost anchor;
- 2 wrong root reference;
- 3 position confinement;
- 4 string-pair confusion;
- 5 octave displacement;
- 6 pattern boundary failure;
- 7 shape copied without interval awareness;
- 8 and failure to connect adjacent regions.

Feedback should identify the missing anchor or relationship rather than simply highlight the destination. Corrected work should require the learner to relocate from another starting point where appropriate.

17. Physical-Execution Errors

A physical-execution error occurs when the learner understands the target but cannot yet perform the required movement reliably.

Subclasses include:

- 1 fretting inaccuracy;
- 2 picking or plucking mismatch;
- 3 coordination failure;
- 4 transition breakdown;
- 5 unnecessary motion;

- 6 muting failure;
- 7 articulation error;
- 8 excessive tension;
- 9 inconsistent tone;
- 10 and tempo-induced breakdown.

Feedback must avoid medical diagnosis. It should describe observable movement or sound, reduce difficulty, isolate the transition, and require controlled repetition.

18. Temporal-Control Errors

A temporal-control error concerns pulse, subdivision, duration, attack placement, rest, accent, synchronization, or tempo stability.

Subclasses include:

- 1 pulse loss;
- 2 subdivision confusion;
- 3 rushing;
- 4 dragging;
- 5 displaced attack;
- 6 shortened or lengthened duration;
- 7 rest omission;
- 8 recovery failure after error;
- 9 click-following without internal subdivision;
- 10 and tempo beyond current control.

Feedback should identify where the note falls relative to the pulse and prescribe counting, tapping, reduced tempo, isolated rhythm, or less frequent click support as appropriate.

19. Musical-Application Errors

A musical-application error occurs when component knowledge is available but is not used appropriately in musical context.

Subclasses include:

- 1 chord-tone selection failure;
- 2 unresolved phrase intention;
- 3 context-inappropriate note use;
- 4 inability to adapt a known pattern;
- 5 loss of time during choice;
- 6 technically correct but functionally irrelevant response;
- 7 and overconstraint that prevents meaningful choice.

Feedback should refer to the musical purpose and constraints, not merely one preferred answer. Open-ended work may contain unconventional but valid choices and must not be marked incorrect solely for differing from a model.

20. Metacognitive-Practice Errors

A metacognitive-practice error occurs when the learner selects or maintains an ineffective practice strategy.

Subclasses include:

- 1 vague target;
- 2 inappropriate tempo;
- 3 scope too large;
- 4 repetition without correction;
- 5 failure to isolate;
- 6 premature difficulty increase;
- 7 inaccurate self-evaluation;
- 8 excessive session continuation after attention declines;
- 9 avoidance of weak areas;
- 10 and review choices based only on preference or familiarity.

Feedback should help the learner make the decision rather than permanently make every decision for them.

21. Attention and Regulation Errors

An attention or regulation error occurs when the learner possesses relevant capability but fails to apply it consistently because of rushing, divided attention, fatigue, frustration, or careless continuation.

Classification requires caution. A system **SHOULD NOT** infer inattention merely because an error appears simple.

Evidence may include:

- inconsistent performance on already stable material;
- immediate self-recognition;
- improvement when the task is repeated without additional instruction;
- or deterioration across a prolonged session.

The response may involve pause, reset, reduced quantity, or rest rather than additional explanation.

22. Transfer Errors

A transfer error occurs when a learner succeeds in practiced conditions but fails when a relevant feature changes.

Subclasses include:

- 1 root dependence;
- 2 position dependence;
- 3 representation dependence;
- 4 key dependence;
- 5 tempo dependence;
- 6 context dependence;
- 7 exact-example memorization;
- 8 and inability to integrate with another skill.

Feedback should identify which incidental feature had become a hidden support and design a near-transfer step before demanding broader variation.

23. Task and System Errors

Not every unsuccessful result is a learner error.

Task and system errors include:

- 1 ambiguous instructions;
- 2 incorrect answer key;
- 3 invalid notation or diagram;
- 4 audio mismatch;
- 5 inaccessible response mode;
- 6 latency or technical failure;
- 7 inconsistent terminology;
- 8 misleading feedback;
- 9 and an assessment that does not measure the claimed capability.

When task validity is compromised, the attempt **MUST NOT** be used as negative evidence against the learner.

24. Error Severity

Severity describes educational consequence, not emotional judgment.

24.1 Minor Error

An isolated, non-systematic mismatch that does not substantially disrupt the target.

24.2 Material Error

An error that meaningfully weakens the current objective or requires correction before continuation.

24.3 Blocking Error

An error that prevents valid performance, understanding, or interpretation of the task.

24.4 Cascading Error

An error in a prerequisite or early step that creates several later mismatches.

24.5 Safety-Relevant Error

An observable practice condition that may reasonably warrant stopping or reducing physical demand, without making medical claims.

Feedback intensity and interruption SHOULD be proportionate to severity.

25. Error Frequency and Pattern

Error interpretation must account for recurrence.

25.1 Isolated Error

Appears once within otherwise stable performance.

25.2 Intermittent Error

Appears inconsistently under similar conditions.

25.3 Patterned Error

Recurrs under identifiable conditions.

25.4 Systematic Error

Reflects a stable misconception, procedure, or execution pattern.

25.5 Generalized Error

Appears across multiple related contexts and may indicate a broader prerequisite weakness.

Repeated patterned or systematic errors MUST trigger a change in instructional response.

26. Feedback Timing

26.1 Immediate Feedback

Immediate feedback is preferred when continued error would strengthen an incorrect physical, rhythmic, or conceptual pattern, or when the learner cannot interpret the task without correction.

26.2 Delayed Feedback

Delayed feedback MAY be used to preserve complete performance, encourage self-detection, assess independent recovery, or avoid fragmenting musical flow.

26.3 Summary Feedback

Summary feedback may follow a set of attempts and should identify the most important pattern rather than list every minor imperfection.

26.4 Learner-Requested Feedback

Learners may sometimes choose when to receive feedback, provided that this does not permit harmful or systematically inaccurate repetition.

Timing decisions must reflect the objective, error severity, and risk of reinforcing the error.

27. Feedback Specificity

Feedback should be specific enough to guide action but not so detailed that it overwhelms the learner or removes all problem solving.

A useful specificity ladder is:

- 1 outcome only;
- 2 dimension identified;
- 3 location identified;
- 4 relationship or rule identified;
- 5 strategy suggested;
- 6 partial cue supplied;
- 7 worked correction supplied;
- 8 complete model supplied.

FretGarden SHOULD begin at the lowest level likely to restore productive work and escalate only when evidence shows it is insufficient.

28. Feedback Tone

Feedback MUST be calm, direct, respectful, and technically precise.

It MUST NOT:

- shame;
- ridicule;
- imply lack of talent;
- exaggerate failure;
- use false praise;
- manufacture urgency;
- infantilize;
- or frame review as punishment.

Encouragement should refer to observable behavior, such as accurate correction, patient tempo reduction, improved consistency, or effective self-diagnosis.

29. Feedback Content Order

When several elements are useful, feedback SHOULD generally follow this order:

- 1 identify what remains correct;
- 2 identify the primary discrepancy;
- 3 state the smallest useful correction;

- 4 define the next attempt;
- 5 and explain the broader principle only if needed.

This order preserves successful components and prevents the learner from reconstructing the entire task unnecessarily.

30. Corrective Feedback

Corrective feedback must produce an action.

Permitted corrective actions include:

- retrying the same item;
- explaining the reasoning;
- selecting between contrasted examples;
- reconstructing from a cue;
- reducing tempo;
- isolating a transition;
- counting or tapping;
- locating an anchor;
- hearing the contrast again;
- or applying the corrected concept in a new example.

Showing the correction without requiring learner action is re-exposure, not complete corrective practice.

31. The Correction Loop

A compliant correction loop contains:

- 1 error observation;
- 2 error localization;
- 3 appropriate feedback;
- 4 learner reconstruction or adjustment;
- 5 reattempt;
- 6 verification;
- 7 prompt reduction where support was added;
- 8 and later retrieval or variation where the objective is core.

The loop should stop when evidence is sufficient, attention is deteriorating, or remediation is required.

32. Reattempt Rules

A reattempt should test the corrected capability, not merely short-term copying.

After complete answer revelation, an immediate identical response is weak evidence. A stronger sequence may include:

- 1 correct the original item;
- 2 explain or identify the relevant relationship;
- 3 complete a near variation;
- 4 return to the original or equivalent item after intervening work;
- 5 and review again after delay.

Unlimited retries without instructional change MUST NOT be treated as learning.

33. Hint Architecture

Hints should be ordered from least revealing to most revealing.

A typical sequence is:

- 1 restate the target;
- 2 direct attention to the relevant dimension;
- 3 identify the region or step;
- 4 provide a strategic question;
- 5 supply an anchor;
- 6 show a partial relationship;
- 7 demonstrate the process;
- 8 reveal the complete answer.

The appropriate sequence varies by domain. Hint use MUST be recorded as part of the evidence condition.

34. Feedback and Guessing

Correct answers reached through guessing require different feedback from confident retrieval.

Where guessing is plausible, FretGarden SHOULD request one or more of the following:

- explanation;
- construction;
- confidence judgment;
- second equivalent item;
- response without alternatives;
- or application.

Feedback MUST NOT praise a lucky selection as though it demonstrated stable knowledge.

35. Feedback and Partial Success

Feedback should preserve dimensions that succeeded.

Examples include:

- correct notes but unstable rhythm;
- correct rhythm but incorrect note;
- correct interval formula but wrong spelling;
- correct location with excessive cue dependence;
- correct self-diagnosis after a failed performance;
- or musically coherent application that missed one specified constraint.

Dimensional feedback prevents unnecessary re-teaching and supports precise practice.

36. Feedback and Self-Correction

Self-correction is a constitutional goal.

FretGarden SHOULD gradually teach learners to ask:

- 1 What was the target?
- 2 What changed from the target?
- 3 Where did the change occur?
- 4 Was the cause conceptual, perceptual, physical, temporal, or strategic?
- 5 What variable should be reduced or isolated?
- 6 What would count as a corrected attempt?

Feedback should increasingly invite learner diagnosis before supplying it.

37. Confidence and Uncertainty

Feedback systems MUST distinguish observation from inference.

Defensible language includes:

- "The third note arrived before the click.";
- "This pattern suggests the root may be misidentified.";
- "You may be relying on the visible shape; try rebuilding it from the interval formula.";
- or "There is not enough evidence yet to determine whether this was a recall error or an accidental selection."

Feedback MUST NOT state an inferred cause as certain when the available evidence does not support certainty.

38. Escalation to Remediation

Local feedback should escalate when:

- 1 the same material error persists after appropriate correction;
- 2 the learner cannot begin without complete support;
- 3 a prerequisite misconception is detected;
- 4 performance collapses across several related examples;
- 5 the learner repeatedly guesses;
- 6 added cues do not improve understanding;
- 7 or the task remains outside productive difficulty.

Escalation may route the learner to prerequisite review, a reduced-scope lesson, a contrasting example, slower practice, or re-teaching. The detailed remediation architecture is governed by Chapter 9.

39. Feedback and Evidence Updates

Feedback changes the conditions under which evidence is produced.

A corrected response after a hint may support guided or reconstructed capability. It **MUST NOT** be recorded as independent capability unless the support is later removed and the learner succeeds.

Evidence records **SHOULD** preserve:

- original attempt;
- error class;
- feedback level;
- corrected response;
- support remaining;
- self-correction status;
- and later independent performance.

40. Feedback in Assessment

Assessment feedback must reflect the purpose of the assessment.

40.1 Formative Assessment

May provide immediate diagnostic guidance and correction.

40.2 Summative Assessment

May delay detailed feedback until evidence collection is complete to avoid contaminating later items.

40.3 Diagnostic Assessment

Should provide error-class information and route to targeted work.

40.4 Retention Assessment

Must avoid supplying enough information to reconstruct later items before retention has been measured.

40.5 Transfer Assessment

Feedback should distinguish failure of the underlying capability from failure to adapt to the changed context.

41. Feedback in Open-Ended Work

Feedback on improvisation, composition, interpretation, and other creative work must distinguish constraints from preferences.

A response is not incorrect merely because it differs from a model.

Feedback may address:

- whether required musical constraints were satisfied;
- whether choices were intentional;
- whether the target concept was used;
- temporal and technical execution;
- coherence;
- and opportunities for revision.

Subjective preference **MUST NOT** be presented as constitutional correctness.

42. Feedback Across Objective Domains

42.1 Perception

Use audible contrast, repeated listening, focused attention, and reproduction.

42.2 Retrieval

Use cue ladders, reconstruction, cue fading, and later independent return.

42.3 Conceptual Reasoning

Use questions, derivation, comparison, and correction of the learner's model.

42.4 Fretboard Navigation

Use anchors, interval relationships, position connections, and varied starting points.

42.5 Physical Execution

Use observable movement, sound, tempo reduction, isolation, and controlled repetition.

42.6 Temporal Control

Use pulse location, subdivision, counting, tapping, and reduced tempo.

42.7 Musical Application

Use function, constraints, choice, and context.

42.8 Metacognitive Practice

Use reflection, strategy comparison, and learner-selected adjustment.

43. Learner-Facing Error Language

Preferred language describes the attempt and next action.

Examples:

- "The note choice was correct, but it arrived one eighth note early. Clap the rhythm once, then replay it at the same tempo.";
- "You found the shape from the root marker. Now rebuild it with the marker removed.";
- "Your chord contains the correct root and fifth, but the third is minor. Raise the third by one fret and compare the sound.";
- "This may be a recall lapse rather than a new concept problem. Try the first-letter cue, then answer again without it.";
- or "You corrected the shift without stopping. Repeat once more to confirm that the recovery is stable."

Discouraged language includes "bad," "easy mistake," "you should know this," "careless" without evidence, and exaggerated celebration after routine correction.

44. Feedback Quantity

More feedback is not always better.

Excessive feedback can:

- divide attention;
- create dependence;
- obscure the primary objective;
- reduce self-detection;
- and make practice feel like continuous interruption.

FretGarden SHOULD prioritize one primary correction at a time during early learning. Additional dimensions may be addressed after the blocking issue is stabilized.

45. Feedback Fading

External feedback should decrease as learner capability increases.

A typical progression is:

- 1 immediate explicit correction;
- 2 immediate location cue;
- 3 strategic question;
- 4 delayed prompt for self-evaluation;
- 5 learner-initiated correction;
- 6 independent monitoring;
- 7 and post-performance reflection.

Permanent immediate correction can prevent the learner from developing internal monitoring.

46. Feedback and Motivation

Feedback may support motivation by making progress and next steps intelligible.

It MUST NOT manipulate learners through shame, fear of lost status, false urgency, or exaggerated reward.

Accurate encouragement may recognize:

- persistence with adjustment;
- willingness to slow down;
- improved error detection;
- successful transfer;
- stable rhythm;
- reduced support;
- or honest self-assessment.

The purpose is to reinforce effective learning behavior, not mere compliance or activity volume.

47. Feedback Anti-Patterns

The following practices are noncompliant or require exceptional justification:

- 1 displaying only "correct" or "incorrect" for a core learning task;
- 2 revealing the full answer after every first error;
- 3 allowing unlimited guessing until success;
- 4 using praise unrelated to observable performance;
- 5 giving several unrelated corrections at once;
- 6 treating all wrong answers as the same error;
- 7 inferring motivation or effort from one response;
- 8 repeating the same failed task without changing support;
- 9 treating immediate corrected performance as mastery;
- 10 using speed as the only feedback dimension;
- 11 correcting creative choices that satisfy the stated constraints;
- 12 blaming the learner for an invalid task;
- 13 hiding contradictory evidence behind a single score;
- 14 and providing feedback without a meaningful next attempt.

48. Constitutional Non-Negotiables

- 1 Core practice MUST provide actionable feedback or a valid method of self-evaluation.

- 2 Feedback MUST align with the stated objective.
- 3 Feedback MUST distinguish observed evidence from inferred cause.
- 4 Error classification MUST be provisional when evidence is limited.
- 5 A task or system error MUST NOT be counted as negative learner evidence.
- 6 Repeated patterned errors MUST change the instructional response.
- 7 Complete answer revelation MUST be followed by later unsupported retrieval when the objective requires independence.
- 8 Corrected performance MUST be interpreted according to the support provided.
- 9 Feedback MUST preserve relevant partial success.
- 10 Feedback MUST remain respectful and non-shaming.
- 11 Guessing MUST NOT be rewarded as stable knowledge.
- 12 Feedback MUST not make claims stronger than the evidence.
- 13 Physical feedback MUST avoid medical diagnosis.
- 14 Open-ended musical choices MUST be evaluated against stated constraints rather than preference alone.
- 15 Feedback SHOULD develop self-correction and decreasing dependence.
- 16 Core correction loops MUST include a reattempt or equivalent learner action.
- 17 Escalation to remediation MUST occur when local feedback is repeatedly insufficient.
- 18 Learner-facing language MUST identify a productive next step.

49. Conformance Tests

A feedback design conforms only if reviewers can answer:

- 1 What objective and dimensions are being evaluated?
- 2 What evidence is actually observable?
- 3 Which error classes can be distinguished validly?
- 4 Which causes remain uncertain?
- 5 What is the smallest useful correction?
- 6 Does the feedback preserve retrieval or problem solving?
- 7 What must the learner do next?
- 8 How will the system know whether the correction worked?
- 9 How is prompting recorded?
- 10 What happens after repeated failure?
- 11 Does the feedback preserve partial success?
- 12 Does it avoid shame and false praise?
- 13 Could the task itself be invalid?

- 14 Does corrected performance support only the claim justified by its conditions?
- 15 How does feedback fade as independence grows?
- 16 How will a similar error be handled in later review?

50. Tradeoffs and Resolution Rules

50.1 Immediate Correction Versus Musical Continuity

Immediate interruption prevents reinforcement of error but may disrupt complete musical performance. Blocking or habit-forming errors should be interrupted; minor errors may be summarized after the phrase or attempt.

50.2 Specificity Versus Cognitive Load

Detailed feedback can improve precision but overwhelm attention. The primary blocking issue should be addressed first, with secondary issues deferred.

50.3 Guidance Versus Retrieval

More guidance can restore success but reduce evidence of independence. Use the least revealing support that permits productive reconstruction, then fade it.

50.4 Automation Versus Diagnostic Humility

Automated systems can respond quickly but may observe only limited dimensions. They must communicate uncertainty and avoid causal claims beyond available evidence.

50.5 Encouragement Versus Honesty

Encouragement should sustain effort without inflating capability. It should identify observable progress while preserving accurate status.

50.6 Error Correction Versus Learner Agency

Direct correction is efficient, but repeated direct correction can create dependence. Learners should increasingly diagnose and select strategies themselves.

50.7 Consistency Versus Context

A shared taxonomy supports coherence, but domain-specific errors require different responses. Common categories should govern interpretation while specialized chapters define valid subtypes and corrections.

51. Assumptions Made

This chapter assumes:

- 1 objectives and evidence conditions are defined according to Chapter 2;
- 2 lessons and exercises follow Chapters 3 and 4;
- 3 assessments identify their claim limits according to Chapter 5;
- 4 mastery and progression states follow Chapter 6;
- 5 review and retention evidence follow Chapter 7;

- 6 some performance dimensions may not be observable in every context;
- 7 feedback may be delivered through text, sound, demonstration, comparison, or structured learner reflection;
- 8 learners may need different amounts of support;
- 9 corrected attempts can be distinguished from independent attempts;
- 10 and Chapter 9 will define complete remediation architecture.

52. Unresolved Questions

The following matters remain unresolved:

- 1 standard learner-facing labels for error classes;
- 2 default limits on feedback quantity per attempt;
- 3 domain-specific timing rules for interruption;
- 4 confidence thresholds for automated error classification;
- 5 treatment of expressive timing and intentional deviation;
- 6 standards for evaluating tone and articulation automatically;
- 7 accessibility-specific feedback alternatives;
- 8 how learner preferences should influence feedback timing and detail;
- 9 evidence aggregation after self-correction;
- 10 when attention-related classifications are sufficiently justified;
- 11 multilingual terminology and feedback consistency;
- 12 and governance for revising error taxonomies after release.

53. Topics Intentionally Deferred

Deferred topics include:

- 1 remediation architecture;
- 2 adaptive difficulty;
- 3 domain-specific feedback libraries;
- 4 human instructor override;
- 5 automated audio and performance evaluation;
- 6 accessibility accommodations;
- 7 learner-controlled feedback settings;
- 8 progress visualization;
- 9 affective and motivational interventions;
- 10 practice-log interpretation;
- 11 detailed creative-work rubrics;

54. Required Future Cross-References

54.1 Remediation and Adaptive Difficulty Chapter

Must reference error families, escalation triggers, hint architecture, correction loops, and evidence updates.

54.2 Domain-Specific Pedagogy Chapters

Must define domain-valid error subtypes, observable evidence, feedback timing, and corrective strategies without weakening this chapter's limits on inference.

54.3 Educational Analytics Chapter

Must distinguish raw response data, error classifications, confidence, and learner-facing claims.

54.4 Accessibility Chapter

Must define equivalent feedback and correction methods where standard response modes are inaccessible.

54.5 Instructional Voice Chapter

Must reference feedback tone, honesty, respect, specificity, and prohibited language.

54.6 Constitutional Governance Chapter

Must define review and amendment procedures for error taxonomies and feedback standards.

55. Chapter Conformance Statement

A FretGarden feedback system conforms to this chapter only when it observes attempts under known conditions, interprets errors without exceeding the evidence, communicates a specific and respectful next action, requires correction or reattempt where appropriate, and updates capability claims according to the support actually provided.

An error classification conforms only when it is objective-relative, pedagogically actionable, proportionate in certainty, and used to select an appropriate instructional response rather than to judge the learner.

Feedback is successful not when it produces one immediately correct answer, but when it improves the learner's ability to detect, understand, correct, retain, and eventually prevent similar errors with decreasing external support.

End of Chapter 8 - Feedback and Error Classification